

THE UNIVERSITY OF
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Final Report

The Art of Scientific Computing: Project Title: TESTING

Author:
Insert name here

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Faculty of Science
The University of Melbourne

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Subject co-ordinators: A/Prof. Suzie Sheehy & Prof. Christian Reichardt

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Chapter 1

Introduction

1.1 Executive Overview

To-do: summary of project, maybe move to preamble or abstract or similar.

1.2 Topological Data Analysis

Brief introduction to TDA. Main concepts relevant for report, detail and definitions deferred to appendix.

1.3 Chronic Traumatic Encephalopathy

Brief introduction to CTE. Focus on why this, as opposed to any other example, may benefit from TDA.

1.4 Workflow

Chapter 2

Data Processing

2.1 Extraction

Where data came from; some notes on biological source, what it is, etc, and how i practically got it from the source.

The data for this report is an image of a normal brain and a brain with CTE, published by a team from the Boston University CTE Center and Brain Bank in [3, Fig. 1].

Biological details, to sort/explain later, (1) coronal section (what is that), (2) cte brain is from a retired professional (American) football player, (3) normal brain no details.

Image extracted as JPEG from published PDF file.

2.2 Pre-Processing

Cleaning of data done before uploading to Python.

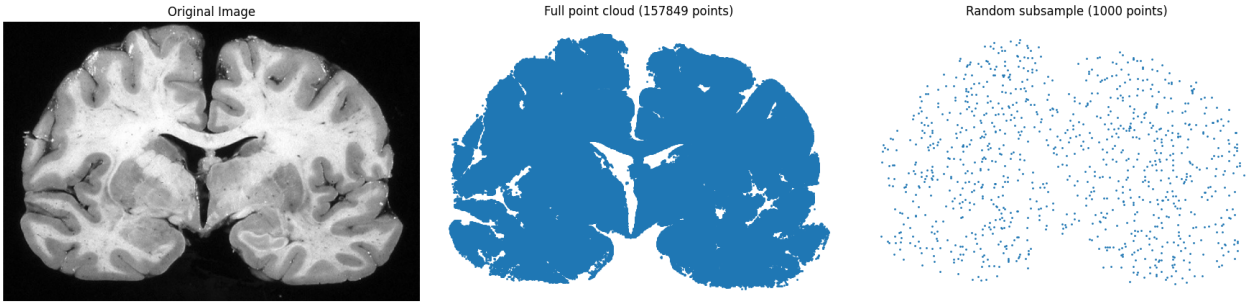
I did two steps of pre-processing in Adobe photoshop.

1. One image of both brains cropped into two images.
2. The authors annotated the CTE brain with some numbers. I used Adobe's wand tool to smoothly remove the annotations.

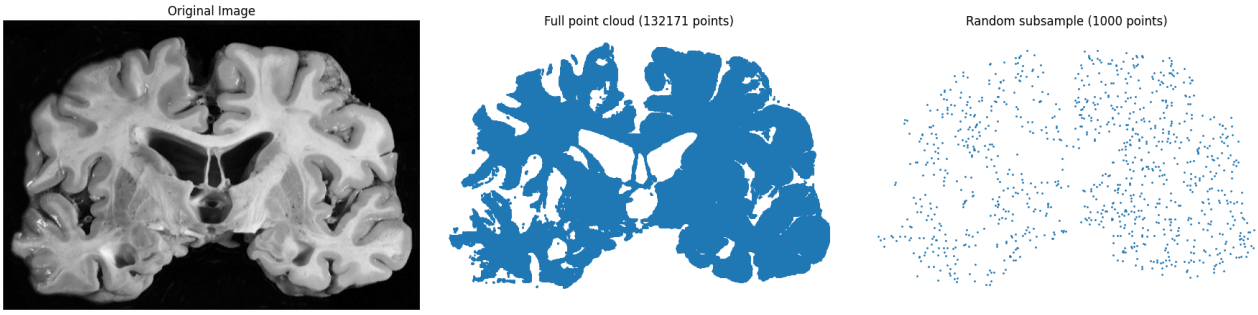
2.3 Processing

Cleaning of data done in Python. Need to justify and explain limitations of the following decisions:

- Converting images to grayscale.
- Convert to point cloud by thresholding pixels by brightness.
- Random subsample of point cloud to fix number of points.



(a) Description of the first plot.



(b) Description of the second plot.

Figure 2.1: Overall caption describing both panels.

Chapter 3

Topological Data Analysis

Intended structure for each section:

- Brief overview of what is being computed on conceptual level, refer to appendix where necessary.
- Overview of algorithm employed by GUDHI.
- Results for each of normal and cte brain.

3.1 Filtrations

GUDHI Tutorial: [1]

Discuss two types of simplicial complexes: VR complex and alpha complex.

Discuss how to tune max edge length parameter.

Choose max dimension of 3 because it is human intuitable.

Brief description of filtrations:

- The control VR complex has 1000 vertices and 24,887,692 simplices.
- The CTE VR complex has 1000 vertices and 33,822,996 simplices.

3.2 Barcodes and Diagrams

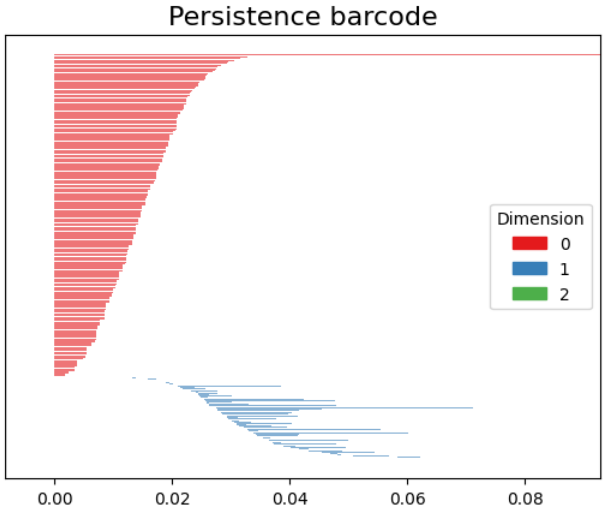
GUDHI Tutorial: [2]

Discuss persistence modules (the mathematical information) and persistence barcodes/diagrams (two equivalent visualisations of persistence modules).

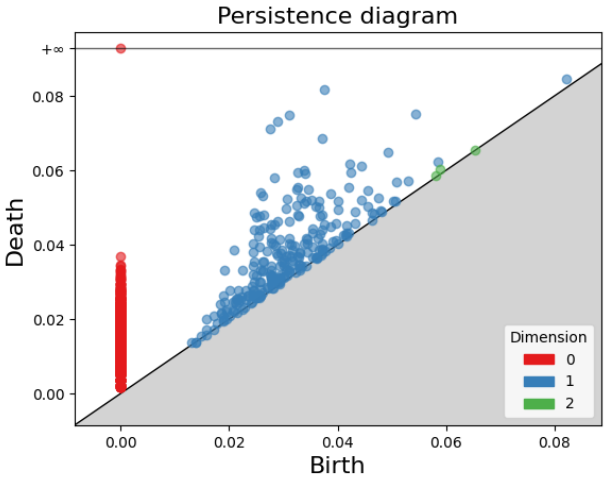
3.3 Comparison

Really, there are two steps to the comparison: (1) are the diagrams/barcodes different? (2) what are the differences?

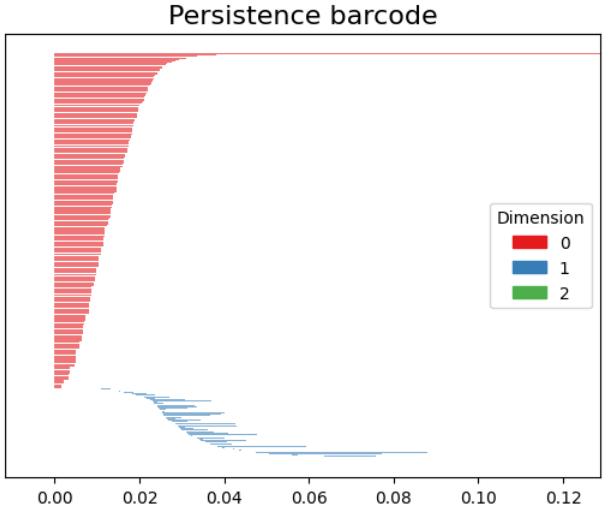
For bare comparison: bottleneck distance vs wasserstein distance, dimension by dimension comparison, interpreting units of measurement, 'significance testing' via subsampling.



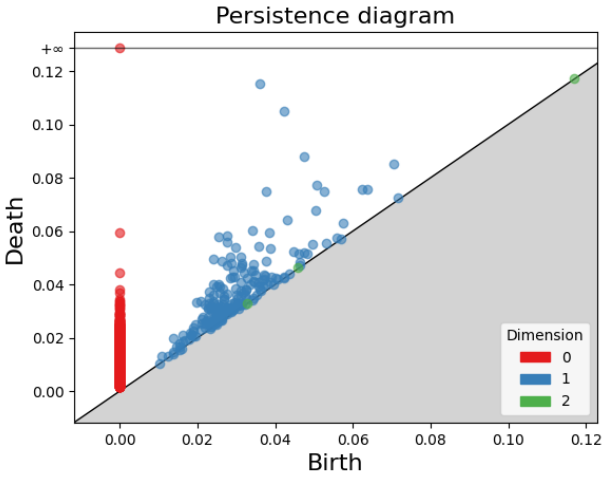
(a) Description of the first plot.



(b) Description of the second plot.



(c) Description of the third plot.



(d) Description of the fourth plot.

Figure 3.1: Overall caption describing all four panels.

Chapter 4

Limitations

List of limitations, to be organised and explained later

1. In data pre-processing: removing author annotations via Adobe photoshop introduces exterior bias/noise to image. In the interests of time, I took this approach. An ideal approach would have been to contact the original authors of the paper and ask for a high resolution copy of the image.
2. In data processing: choosing to study images in grayscale rather than colour. What is lost and how would I have approached it with colour?

Appendix A

Mathematics of TDA

Bibliography

[1] F. Chazal and B. Michel. Building simplicial complexes from a point cloud: Rips and alpha complexes. `https://github.com/GUDHI/TDA-tutorial/blob/master/tutorials/Tuto-GUDHI-simplicial-complexes-from-data-points.ipynb`, 2025. GitHub repository. Accessed: 2026-10-14.

[2] G. Project. Persistent homology and persistence diagrams. `https://github.com/GUDHI/TDA-tutorial/blob/master/tutorials/Tuto-GUDHI-persistence-diagrams.ipynb`, 2025. GitHub repository. Accessed: 2026-10-14.

[3] R. A. Stern, D. O. Riley, D. H. Daneshvar, C. J. Nowinski, R. C. Cantu, and A. C. McKee. Long-term consequences of repetitive brain trauma: chronic traumatic encephalopathy. *PM&R: The Journal of Injury, Function, and Rehabilitation*, 3(10 Suppl 2):S460–S467, Oct. 2011.